ServiceNow Application Development

Project Documentation – Employee Raise Issues

Project	Employee Raise Issues (Employee Issue Management System)
Platform	ServiceNow – custom application with a Service Portal front end
Application	Employee Center Custom
Custom table	Employee raises issue (x_2107363_employ_0_employee_raises_issue)
Record Producer	Raise Employee Issue
Date	22 October 2026

1. Introduction

- **Project Title:** Employee Raise Issues (Employee Issue Management System)
- **Team Members:** List team members and their roles.

2. Project Overview

- **Purpose:** Employees often report workplace problems through scattered emails, phone calls and informal messages, so requests get lost and nobody can see their progress. Employee Raise Issues solves this with one Service Portal where employees submit an issue through a simple form. HR and IT teams then receive, assign and resolve it, and the employee can follow the issue to the end.
- **Objectives:**
  - Make issue submission quick, with no need to open backend forms.
  - Capture the same key details every time: requester, category, subcategory, short description and attachments.
  - Store every issue in one custom table with a unique number.
  - Route issues to the right support group and track state until closure.
  - Limit access by role, so each user sees only what they should.
- **Features:**
  - Custom application and table, with auto-numbered records (prefix EMP, for example EMP0001009).
  - Record Producer “Raise Employee Issue” that creates records in the custom table.
  - Service Portal home page with a “Raise an Issue” card and a Raise Ticket button.
  - Dependent form behaviour through UI Policies, and file attachments with drag and drop.
  - Save as Draft and Submit options on the portal form.
  - Issue lifecycle from New to Closed, with assignment groups and resolution notes.
  - Role- and ACL-based security, and optional notifications through Flow Designer.

Issue lifecycle (configure actual state values in your instance):



3. Architecture

End-to-end flow:



- **Frontend:** Service Portal. The home page shows the "Our Commercial Services" panel (24/7 Customer Support, Faster Issue Resolution, Transparent Tracking, Secure & Reliable) and the "Raise an Issue" card. Raise Ticket opens the Raise Employee Issue form through the Service Catalog (Home > Service Catalog > Office > Services).
- **Backend:** Custom application "Employee Center Custom" built in Studio. The Record Producer writes into the custom table, and UI Policies control form behaviour. A Record Producer script runs before the record is generated, and the user is redirected to the generated record after submission. Assignment rules, Flow Designer flows, roles and ACLs handle routing, notifications and security.
- **Database:** The table "Employee raises issue" holds one record per issue. Its columns, as defined in the instance, are listed below.

Column	Type	Status	Purpose
Number	String (auto-number)	Implemented	Unique issue ID, default getNextObjNumberPadded()
Requester	Reference → User	Implemented	Employee who raised the issue
Category	String	Implemented	Network, Hardware, Software or Access
Subcategory	String	Implemented	VPN, Laptop, Server or Forgot password
Short description	String	Implemented	Brief summary of the issue
State	String	Implemented	Current progress of the issue
Priority	As configured	On form	Urgency of the issue
Assignment Group	Reference → Group	Implemented	Responsible support team
Created / Updated / Updated by / Updates	Date-Time, String, Integer	System	Audit information kept by the platform
Description	String (long text)	Recommended	Detailed explanation of the problem
Assigned To	Reference → User	Recommended	Person handling the issue
Resolution Notes	String (long text)	Recommended	How the issue was resolved

4. Setup Instructions

- **Prerequisites:**
  - A ServiceNow instance (for example a Personal Developer Instance) with the admin role.
  - Access to Studio, Service Catalog, Service Portal and Flow Designer.
  - Test users, plus HR/IT support groups for assignment testing.
- **Installation:** Step-by-step guide to build the application.
  1. Open App Engine → ServiceNow Studio and create the application. Confirm it is the active application before adding components.
  2. Create the table "Employee raises issue" with the columns listed in Section 3. Turn on auto-numbering for the Number field.
  3. Create UI Policies that make fields visible, mandatory or read-only for each issue category.
  4. Create the Record Producer "Raise Employee Issue" with the table as its target. Add the variables (requester, category, subcategory, state, short description, attachments) and map them to table columns.
  5. Create or select a Service Portal. Set its title and URL suffix, then add the issue submission page.
  6. Add the Record Producer to the portal page through the catalog component or a configured widget.
  7. Configure assignment, states and notifications, then give support users their roles.

5. Folder Structure

A ServiceNow application has no folder tree. Its components are grouped inside the application, as shown below.

Area	Components
Application (Studio)	Employee Center Custom, the container for everything below
Data	Table "Employee raises issue" with columns, choice lists and dictionary entries
Client-side	UI Policies and Client Scripts for dynamic form behaviour
Service Catalog	Record Producer "Raise Employee Issue": variables, mapping and pre-insert script
Service Portal	Portal record, home page, submission page and widgets
Server-side and security	Assignment rules, Flow Designer flows, notifications, roles and ACLs

6. Running the Application

- **Employee:** Open the Service Portal and log in. Select Raise Ticket on the Raise an Issue card. Complete the Raise Employee Issue form and select Submit. The new record gets the next EMP number.
- **Support user:** Open the Employee raises issue list, open the new record, assign it to a group and update the state. Add resolution notes when the work is done.
- **Administrator:** In the Record Producer, select Try It to preview the form without opening the portal.

7. API Documentation

The project adds no custom REST API. Issues are created through the Record Producer. Records can also be read or written through the standard ServiceNow Table API, subject to ACLs. The table name is x_2107363_employ_0_employee_raises_issue.

Method	Endpoint	Description
GET	/api/now/table/x_2107363_employ_0_employee_raises_issue	List issues (supports sysparm_query and sysparm_limi
GET	/api/now/table/x_2107363_employ_0_employee_raises_issue/{sys_id}	Retrieve one issue
POST	/api/now/table/x_2107363_employ_0_employee_raises_issue	Create an issue. Example body: {"short_description":"VPN not connecting","category":"Network
PATCH	/api/now/table/x_2107363_employ_0_employee_raises_issue/{sys_id}	Update fields such as state or resolution notes

8. Authentication

- **Authentication:** Employees sign in to the Service Portal with their ServiceNow accounts (local, or SSO/LDAP if the instance uses it). The platform manages the session, and the application adds no separate token handling.
- **Authorization:** Roles and ACLs decide who can read, create and update records.
  - Employees: submit issues and view their own issues.
  - Support users (HR/IT): view assigned or group issues, update state and add resolution notes.
  - Administrators: manage the table, Record Producer, portal and configuration.
- **API access:** REST calls use basic authentication or OAuth, and the same ACLs apply.

9. User Interface

- **Portal home page:** The “Our Commercial Services” panel sits beside the “Raise an Issue” card (Submit your request or complaint), which holds the Raise Ticket button.

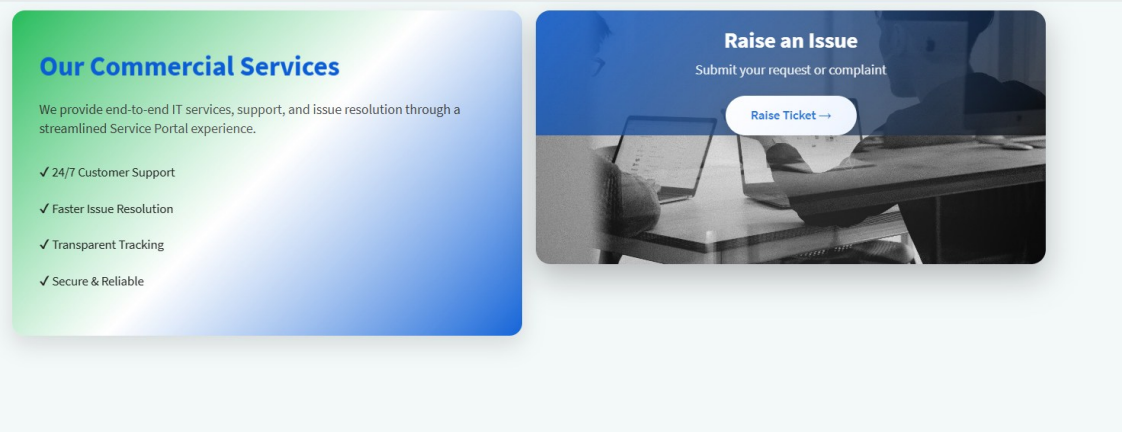


Figure 1: Service Portal home page

- **Raise Employee Issue form:** The Record Producer form shows these fields, with Save as Draft and Submit buttons on the right.
  - Requester name (user lookup)
  - Category: Network, Hardware, Software, Access
  - Subcategory: VPN, Laptop, Server, Forgot password
  - State: New, In progress, Resolved, Closed
  - Short description (text)
  - Add attachments (choose a file or drag and drop)

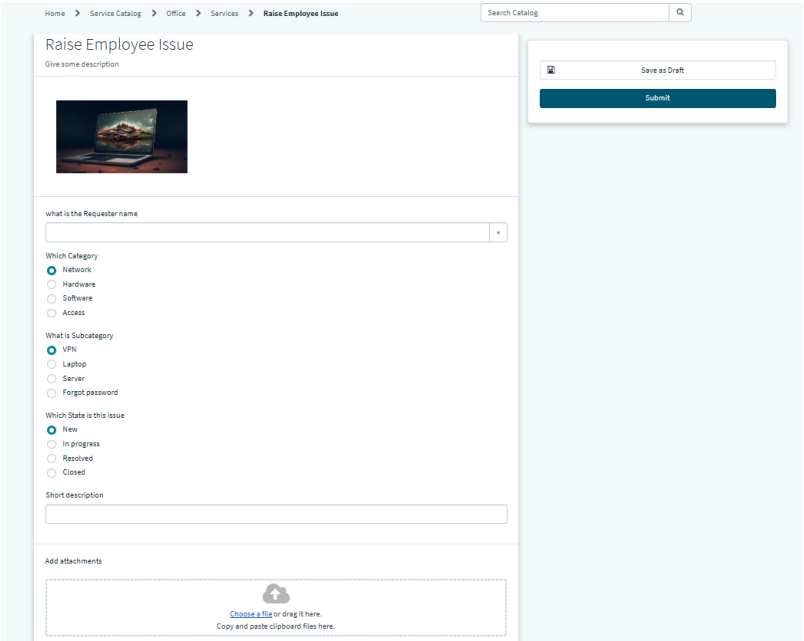


Figure 2: Raise Employee Issue form on the portal

10. Testing

Testing covered the employee's view and the support team's view. Tools used: the portal itself, impersonation of test users, the list and form views, and the system logs. All test cases below were completed.

No.	Test case	Expected result	Status
1	Submit a valid issue through the portal	A record is created in Employee raises issue	Completed
2	Leave a required field empty	Form blocks submission and shows a message	Completed
3	Select each category	UI Policies show or hide the right fields	Completed
4	Compare variables with record fields	Each variable maps to the correct column	Completed
5	Log in as employee, support user and admin	Each sees only what roles and ACLs allow	Completed
6	Assign the issue and change its state	Assignment and state changes are saved	Completed
7	Trigger notification events (if configured)	Notifications reach the right users	Completed
8	Open the portal as an intended employee	Portal loads and the form is usable	Completed

11. Screenshots or Demo

Figures 1 and 2 (Section 9) show the employee-facing portal. The figures below show the back-end configuration.

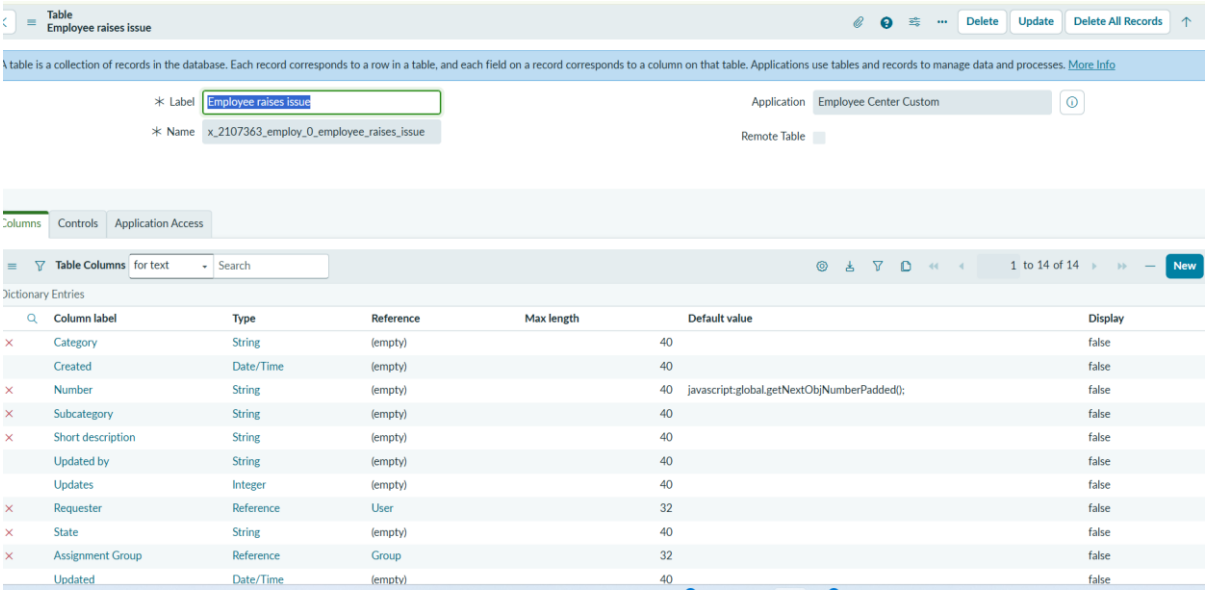


Figure 3: Custom table "Employee raises issue" and its columns

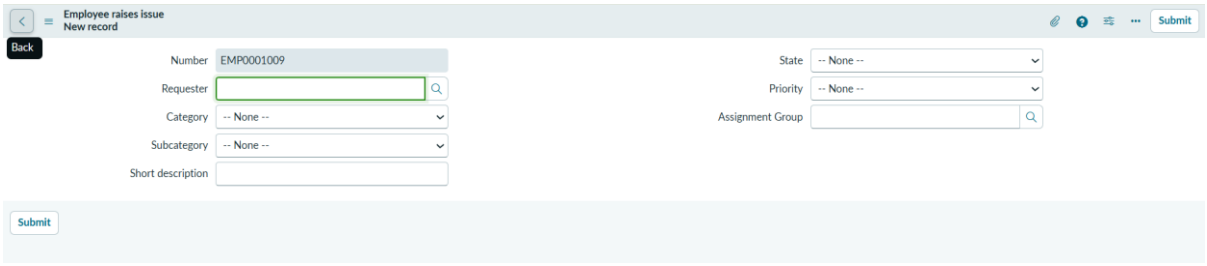


Figure 4: New record form with auto-generated number (EMP0001009)

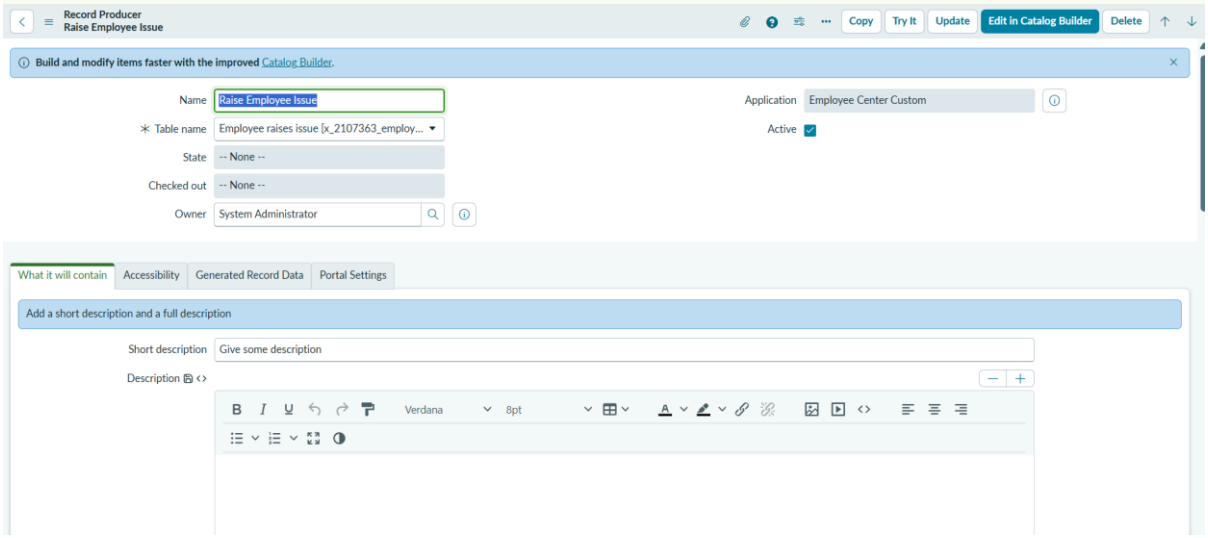


Figure 5: Record Producer "Raise Employee Issue" linked to the custom table

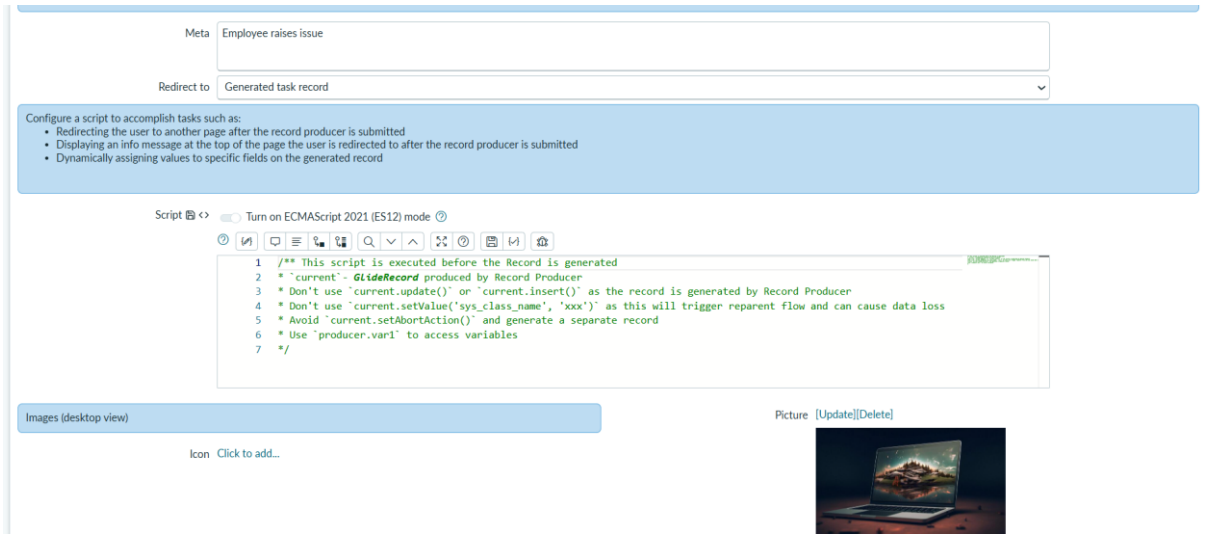


Figure 6: Record Producer redirect setting, script and picture

12. Known Issues

Points noticed in the current configuration that are worth reviewing:

- **Short description length:** The column's maximum length is 40 characters, which is short for a summary. Raise it (the platform default is 160).

- **Free-text fields:** Category, Subcategory and State are String columns, so the database does not enforce valid values. Choice fields would keep data consistent.
- **State on the form:** Requesters can set State, including Resolved or Closed, at submission. Hide it and default it to New.
- **Subcategory filtering:** All subcategories appear under every category (for example Laptop under Network). Filter them with a UI Policy or dependent choices.
- **Placeholder text:** The Record Producer short description still reads “Give some description”. Replace it with real guidance.
- **Application name:** The table belongs to the application “Employee Center Custom”. Confirm this is the scope name you want to present.

13. Future Enhancements

- Email and SMS notifications for submission, assignment and resolution.
- SLA timers and automatic escalation for unresolved issues.
- A “My Issues” portal page where employees track status and add comments.
- A satisfaction survey after an issue is closed.
- Dashboards for issue volume, category trends and resolution time.
- Knowledge-base suggestions while the employee types a description.
- A mobile-friendly view or the Now Mobile app.